

KULLENS FYR, Sweden

WMO: 026060

Lat: 56.3007N

Lon: 12.4520E

Elev: 236

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.4	21.0	6.8	7.8	19.7	12.1	10.3	24.3	54.9	41.6	50.9	42.8	11.8	20	0.840

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	74.7	62.4	71.6	61.5	69.0	60.3	65.0	71.2	63.2	69.2	61.6	67.3	9.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	85.5	68.1	60.5	79.5	66.4	58.9	75.0	64.9	30.1	71.6	28.7	69.0	27.6	67.3	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 46.9	42.0	35.7	18.9	79.5	8.7	4.1	12.6	82.5	7.5	84.9	2.7	87.2	-3.6	90.2		
(5)			16.6	67.1	8.1	2.3	10.8	68.7	6.1	70.1	1.5	71.4	-4.4	73.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	34.1	33.3	36.4	42.9	52.4	58.0	62.5	61.1	55.1	48.2	40.7	36.3		
	DBStd	11.60	6.70	6.33	5.44	6.21	5.63	4.98	4.65	3.83	3.58	4.83	4.51	4.90		
	HDD50	2446	492	466	423	234	33	1	0	0	5	89	278	425		
	HDD65	6681	957	886	888	663	393	217	107	135	296	521	728	890		
	CDD50	1293	0	0	0	22	107	241	388	344	158	33	0	0		
	CDD65	52	0	0	0	0	2	6	30	14	0	0	0	0		
	CDH74	127	0	0	0	0	7	9	89	22	0	0	0	0		
	CDH80	12	0	0	0	0	0	0	10	2	0	0	0	0		
	WSAvg	17.5	22.0	20.2	19.5	14.5	13.6	14.0	14.8	16.3	17.6	18.3	18.5	20.5		
	PrecAvg	25.9	2.0	1.7	1.4	1.5	1.8	2.7	2.5	3.0	2.6	2.4	2.1	2.2		
(15) Precipitation	PrecMax	33.9	3.8	3.4	3.1	2.8	3.6	5.7	5.1	7.2	5.5	4.6	4.0	4.1		
	PrecMin	18.5	0.2	0.2	0.3	0.4	0.7	0.2	0.2	1.1	0.5	1.3	0.6	0.8		
	PrecStd	3.8	1.0	0.7	0.8	0.7	0.9	1.4	1.5	1.6	1.4	0.9	0.8	0.9		
	PrecStd	3.8	1.0	0.7	0.8	0.7	0.9	1.4	1.5	1.6	1.4	0.9	0.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	47.6	50.3	70.3	73.6	75.2	81.2	76.8	67.3	60.1	49.9	47.5		
		MCWB	42.5	45.0	45.0	54.9	60.8	60.4	66.1	63.0	60.3	55.7	47.8	46.1		
	2%	DB	43.1	44.1	46.2	62.6	68.2	71.3	76.2	72.4	63.3	57.3	48.6	45.2		
		MCWB	41.5	41.9	42.0	50.4	55.8	60.7	63.0	62.7	57.2	54.1	46.8	43.8		
	5%	DB	42.1	42.1	44.0	56.8	65.1	68.2	72.7	69.4	61.2	55.6	47.5	43.8		
		MCWB	40.5	40.0	41.1	48.2	54.0	59.8	62.2	62.1	56.2	52.7	45.6	42.3		
	10%	DB	41.1	40.6	42.4	51.2	61.6	65.7	69.9	66.7	59.8	54.2	46.4	42.5		
		MCWB	39.6	38.7	40.1	45.9	52.4	58.8	61.2	60.3	55.2	51.5	44.3	41.1		
	0.4%	WB	43.1	45.6	45.4	54.9	61.3	64.0	68.5	67.1	61.1	57.3	48.6	46.3		
		MCDB	43.7	47.2	49.6	69.3	72.2	69.4	76.5	73.6	65.9	59.4	49.2	47.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.9	42.0	43.1	51.0	57.1	62.7	65.7	64.6	58.9	54.7	47.4	44.1		
		MCDB	42.6	43.6	45.3	59.9	65.2	67.7	72.5	69.9	61.7	56.4	48.3	44.9		
	5%	WB	40.8	40.3	41.7	49.0	55.0	61.1	63.8	62.6	57.4	53.5	45.9	42.6		
		MCDB	41.7	41.8	43.6	56.1	62.8	66.5	70.4	67.9	60.2	55.3	47.0	43.5		
	10%	WB	39.8	39.1	40.4	46.1	53.0	59.4	62.1	60.9	56.0	52.2	44.7	41.2		
		MCDB	41.0	40.4	42.2	50.9	60.1	65.2	68.2	66.1	59.0	54.0	46.0	42.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.3	5.4	7.5	8.7	7.8	7.8	6.6	5.6	5.4	4.3	4.3		
		MCDBR	4.4	4.6	7.2	14.1	14.2	12.1	12.2	9.6	7.5	6.6	4.6	4.5		
		MCWBR	4.6	4.3	5.4	7.9	8.9	7.8	7.3	6.3	5.4	5.6	4.9	4.8		
	5% WB	MCDBR	4.1	4.5	6.6	12.5	12.3	9.8	9.8	8.4	7.0	5.9	4.4	4.3		
		MCWBR	4.4	4.5	5.5	7.7	8.3	6.9	7.1	6.6	6.1	5.5	5.0	4.8		
			4.4	4.5	5.5	7.7	8.3	6.9	7.1	6.6	6.1	5.5	5.0	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.331	0.351	0.385	0.382	0.374	0.397	0.397	0.367	0.350	0.330	0.297		
	taud		2.283	2.317	2.337	2.302	2.378	2.425	2.377	2.382	2.446	2.411	2.326	2.253		
	Ebn at Noon		186	230	256	264	272	276	266	257	250	221	175	157		
	Edh at Noon		17	24	30	36	35	34	35	33	27	22	16	13		
(44) All-Sky Solar Radiation	RadAvg		133	319	752	1269	1654	1743	1617	1300	914	441	166	90		
	RadStd		25	51	77	167	159	147	193	108	102	66	30	10		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (68 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-207	-304	N/A	N/A

Nomenclature: See separate page